

# Ahmad Raza

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## SUMMARY

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Software Engineering Graduate specializing in AI, Machine Learning, and NLP with hands-on experience building multi-agent LLM systems. Research focus on retrieval-augmented generation and cognitive adaptation in educational systems. Adept at driving innovation and generating measurable improvements across diverse projects.

## EDUCATION

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### Thal University Bhakkar

*Bachelor of Science in Software Engineering*

Punjab

Oct. 2022 – May 2026

- CGPA: 3.83/4.0
- Relevant Coursework: Software Engineering, Data Structures & Algorithms, Web Engineering, Database Systems, Cloud Computing

## PROJECTS

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### Sage (FYP) | *LangChain, LangGraph, llama.cpp, FastAPI, React* | [GitHub](#)

Feb–May. 2026

- Built an offline multi-agent academic assistant (8 LangGraph agents) with a dual-model inference; hybrid RAG pipeline (teammate-led) combining dense retrieval and BM25.
- Engineered a hardware-aware llama.cpp lifecycle manager with dynamic context-window scaling (3k–64k), VRAM-based GPU offloading, and reduced server cold-start from ~17s to ~4s via BayesOpt(Optuna)-tuned compiler flags.
- Shipped as a Windows desktop app (pywebview, NSIS installer) across 4 tiered builds (CPU/CUDA × full/lite) with SHA-256 integrity validation on all staged artifacts; CI/CD via GitHub Actions with automated builds upload to Cloudflare R2.
- Enabled zero-code institutional adoption via a TOML-config driven end-to-end pipelines, featuring fully offline, privacy-first deployment in resource-constrained environments.

### Seq Modeling From Scratch | *Python, NumPy, Math* | [GitHub](#)

Nov. 2025

- Implemented RNN, LSTM, GRU without auto-differentiation frameworks to master gating mechanisms and computational graphs.
- Manually derived and coded forward and backward passes (Backpropagation Through Time) to manage gradient flow in dynamic networks and enable training.
- Built Seq2Seq models with Bahdanau and Luong attention mechanisms specifically for language modeling tasks. Trained character and word-level language models to demonstrate practical sequence learning.

### NeuroScope | *Python, NumPy, Matplotlib* | [GitHub](#) | [PyPI](#) | [Docs](#)

Sep 2025

- Engineered a production-grade Python framework for informed training and diagnosis of neural networks.
- Developed a real-time diagnostics system monitoring 10 health indicators (e.g., Dead Neurons, Gradient SNR, Weight Update Ratios) and a comprehensive analytics and evaluation system validated against research by Glorot, He, and Pascanu.
- Designed dynamic visualizations for neuron activation, gradients, and weights histograms and evolution plots to track training dynamics.
- Maintained 60%+ test coverage and automated PyPI publishing workflows via GitHub Actions.

## RESEARCH

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### CLAG: Stateless Cognitive Adaptation for Retrieval-Augmented Generation | *In Preparation*

2026

- Proposed a stateless cognitive adaptation paradigm for educational RAG that estimates a learner's cognitive level from a single query via Bayesian posterior fusion with EM-estimated population priors.
- Derived formal guarantees: viability of single-query adaptation, adaptation monotonicity, and bounded ZPD targeting error under measurable conditions.

## TECHNICAL SKILLS

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**Deep Learning & ML:** LLMs, Transformers, Fine-tuning, RAG, Multi-Agent Systems, Sequence Models, Representation Learning, Statistical Learning Theory, Self-Supervised Learning

**Foundations:** Linear Algebra, Multivariate Calculus, Probability Theory, Statistics, Optimization

**MLOps & Infra:** FastAPI, MLflow, DVC, Model Deployment & Serving, Docker, CI/CD

**Languages & Libraries:** Python, SQL, NumPy, Matplotlib, Seaborn, Scikit-Learn, XGBoost, Optuna, PyTorch, LangChain, LangGraph

## CERTIFICATIONS

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**DeepLearning.AI (Coursera):** Deep Learning Specialization, Machine Learning Specialization, Mathematics for Machine Learning & Data Science Specialization

**Other Specializations:** MLOps Specialization (Duke University), IBM Machine Learning Professional Certificate